

# Exam 2

個人成績報告

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逐題總覽

| 題號  | 題型  | 狀態    |
|-----|-----|-------|
| Q1  | 簡答題 | ✓     |
| Q2  | 簡答題 | ✓     |
| Q3  | 簡答題 | ✓     |
| Q4  | 簡答題 | ✓     |
| Q5  | 簡答題 | ✓     |
| Q6  | 簡答題 | ✓     |
| Q7  | 簡答題 | — 未作答 |
| Q8  | 簡答題 | ✓     |
| Q9  | 簡答題 | ✓     |
| Q10 | 簡答題 | ✓     |
| Q11 | 簡答題 | ✓     |
| Q12 | 簡答題 | ✓     |
| Q13 | 簡答題 | ✓     |
| Q14 | 簡答題 | ✓     |
| Q15 | 簡答題 | ✓     |
| Q16 | 簡答題 | ✓     |
| Q17 | 簡答題 | ✓     |
| Q18 | 簡答題 | ✓     |
| Q19 | 簡答題 | ✓     |
| Q20 | 單選題 | ✓     |
| Q21 | 單選題 | ✓     |
| Q22 | 單選題 | ✓     |
| Q23 | 單選題 | ✓     |
| Q24 | 單選題 | ✓     |
| Q25 | 單選題 | ✓     |
| Q26 | 單選題 | ✓     |
| Q27 | 單選題 | ✓     |
| Q28 | 單選題 | ✓     |
| Q29 | 單選題 | ✓     |
| Q30 | 單選題 | ✓     |

| 題號  | 題型  | 狀態 |
|-----|-----|----|
| Q31 | 單選題 | ✓  |
| Q32 | 單選題 | ✓  |
| Q33 | 單選題 | ✓  |
| Q34 | 單選題 | ✓  |
| Q35 | 單選題 | ✓  |
| Q36 | 單選題 | ✓  |
| Q37 | 單選題 | ✓  |
| Q38 | 單選題 | ✓  |
| Q39 | 單選題 | ✓  |
| Q40 | 單選題 | ✓  |
| Q41 | 單選題 | ✓  |
| Q42 | 單選題 | ✓  |
| Q43 | 單選題 | ✓  |
| Q44 | 單選題 | ✓  |
| Q45 | 單選題 | ✓  |

## 題目詳情

### Q1 簡答題 (3分)

What are the three fundamental exceptions that can occur during message exchanges in a client-server model?

你的答案

LostReq：請求訊息遺失。

LostResp：回應訊息遺失。

SerCrash：請求執行失敗。

### Q2 簡答題 (3分)

Describe the three main types of redundancy used to achieve fault tolerance.

你的答案

1.資訊冗餘：加入額外資訊(如ECC)來偵測或修正錯誤。

2.時間冗餘：將同一操作重複執行多次，以確認結果正確或抵抗暫時性錯誤。

3.實體冗餘：使用多個硬體元件(如多台伺服器或多個CPU)來執行相同任務，以提昇系統與容錯能力。

### Q3 簡答題 (2分)

Identify two major differences between a traditional Multiprocessor (MP) and a Chip-level Multiprocessor (CMP).

你的答案

- 1.硬體整合方式：Normal MPs是多個獨立CPU(多晶片)透過匯流排或網路連接，而Chip-level MPs則是多個核心整合在同一顆晶片上。
- 2.通訊延遲與效率：Normal MPs須透過外部的BUS通訊，延遲較高且成本較大，而CMP在晶片內部通訊(on-chip)，則延遲較低，效率較高。

### Q4 簡答題 (3分)

How does affinity scheduling impact cache misses, TLB misses, and page faults in a multiprocessor system?

你的答案

親和性排程讓process盡量在啊一CPU上執行，因此可以降低TLB misses。因為該CPU的TLB仍保留該process的地址轉換資訊，但它無法降低page faults是由資料是否存在於主記憶體決定，與是否在同一CPU執行無關。

### Q5 簡答題 (3分)

Define Access Transparency, Location Transparency, and Replication Transparency in distributed systems.

你的答案

- 存取透明度：讓使用者能以相同的操作方式來存取本地或遠遠端的訊息資源。
- 位置透明度：讓使用者無需知道資源的實際位置即可存取它們。
- 複製透明度：允許使用者多個資源副本來提昇系統的可靠性與效能，而使用者或應用程式無需知道這些副本的存在。

### Q6 簡答題 (2分)

Compare the number of switches required for a crossbar network versus an Omega network for  $n$  CPUs and  $n$  memory modules.

你的答案

Crossbar :  $N^2$   
Omeage :  $n/2 \log_2 n$

### Q7 簡答題 (4分)

List four methods to notify a receiver in a non-blocking message passing system that a message has arrived.

#### Q8 簡答題 (3分)

Compare Sender-Initiated (SI-LB) and Receiver-Initiated (RI-LB) Load Balancing algorithms. Which one maintains performance under heavy load?

你的答案

發送端主導負載平衡：是指當某個節點負載過重時，主動將工作(tasks)推送(push)到其他節點，適合系統整體負載較低時使用，因為較容易找到空閒節點。而Receiver-Initiated(接收端主導負載平衡)則是當某個節點空閒時，主動向其他節點拉取(pull)工作，適合在系統負載較高使用，因為較容易找到忙碌節點來分擔負載。

#### Q9 簡答題 (2分)

While using the Test-and-Set-Lock (TSL) instruction for mutex locks in MPs may cause system bus contention, please suggest two ways to mitigate it.

你的答案

- 1.在TSL指令之前先以普通Read檢查lock，再必要時才使用TSL讓CPU在本地cache中忙等，避免頻繁鎖住bus。
- 2.使用backoff機制，當取得lock失敗時延遲一段時間再重試，減少多個CPU同時競爭bus的情況。

#### Q10 簡答題 (2分)

In multicasting systems, specify the meaning of 0-reliable and M-out-of-N-reliable semantics.

你的答案

- 0級可靠性：不期望任何接收方回應，只單純單向發送(如時間廣播)。
- M-out-of-N：共有N個接收者，只需其中有M個回應即可( $1 < M < N$ )，常用於容錯與冗餘系統(如RAID)。

#### Q11 簡答題 (2分)

Explain how a channel-based publish/subscribe system can be implemented using group communication.

你的答案

在channel-based中的publish-subscribe系統中，每一個channel可以對應到一個群組。發布者將訊息送到該channel，實際上就是對該群組進行群播；而訂閱者則透過加入或離開對應的群組來表達訂閱或取銷訂閱。如此一來，群組通訊服務負責將訊息自動分發給所有群組成員，從而實現channel-based的發布與訂閱。

**Q12 簡答題** (2分)

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In an SMP system, under what conditions is it more appropriate to use a spin lock versus a blocking lock?

你的答案

在SMP系統中提指執行緒持續在CPU上忙等檢查鎖是否可用，不進行context switch，因此延遲低但會消耗CPU資源，適合用於臨界區很短或預期等待時間很短的情況。

**Q13 簡答題** (3分)

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What are the three basic architectural approaches for Multiprocessor Operating Systems?

你的答案

- 1.每個CPU各自有各自的作業系統。
- 2.主從式多處理器(Master-Slave MP)，由一個Master CPU控制系統，其餘Slave CPU執行任務。
- 3.對稱式多處理器(SMP)所有CPU平等，共同執行同一作業系統並共享資源。

**Q14 簡答題** (4分)

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List four common failures that may happen during a Remote Procedure Call (RPC).

你的答案

- 1.無法找到伺服器/ 伺服器當機/版本不匹配/其他各種錯誤。
- 2.請求訊息遺失。
- 3.回覆訊息遺失。
- 4.伺服器在收到請求後，在處理後或處理中崩潰。
- 5.用戶端發送請求後當機。

**Q15 簡答題** (2分)

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Explain the Cache Coherence problem and the basic rules a system must follow when a CPU wants to write a cached word.

你的答案

在多處理器系統中，當同一筆資料複製到多個CPU的Cache中時，如果某一個CPU修改了該筆資料值時，其他的Cache中副本可能仍是舊值，而導致資料不一致問題，因此系統必須透過一致性協議來確保所有對應該資料的內容保持一致性。

**Q16 簡答題** (3分)

State three major advantages of the Publish/Subscribe middleware model over Point-to-Point communication.

你的答案

- 1.解耦合：Publish與Subscribe不需要知道彼此的存在或位置(時間、空間、同止解耦)提升系統彈性。
- 2.可擴展性：支援一對多、多對多通訊，新增或移除節點不需修改既有程式。
- 3.彈性/動態訂閱：subscribe可動態加入或離開，並依需求訂閱特定事件或主題。

**Q17 簡答題** (3分)

Demonstrate how a counting semaphore initialized to N can be implemented using the Linda coordination language.

你的答案

1. 初始化：執行out("sem") N次。
- 2.wait():進入"sem"狀態。
- 3.signal()：退出"sem"狀態。

**Q18 簡答題** (4分)

Provide two advantages and two disadvantages of a Master-Slave architecture for multiprocessors.

你的答案

優點：1.設計簡單：由master CPU負責排程與控制，其餘Slave CPU只執行任務。2.容易管理與同步，所有關鍵操作集中在Master，以避免複雜的同止問題。  
缺點：效能瓶頸，所有請求都需master，容易成為瓶頸。2.單點故障，一旦master故障，則整個系統可能無法操作。

**Q19 簡答題** (2分)

Why are non-blocking operations less prone to deadlock?

你的答案

發送者與接收者不需互相等待即可繼續執行，提高系統並行度與效能，並且較不容易發生deadlock狀態。

**Q20** 單選題 (2分)

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Gang scheduling in multiprocessor systems is primarily utilized to solve which of the following issues?

- A. High page fault rates
- B. TLB thrashing
- C. Out-of-phase communication among related threads
- D. Excessive bus contention

你的選擇

你的答案: **D**

**Q21** 單選題 (2分)

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Which of the following interconnection network topologies stalls the entire chain of flits across multiple switches if the packet header cannot proceed due to a busy output channel?

- A. Circuit Switching
- B. Store and Forward Switching
- C. Cut-Through Switching
- D. Wormhole Routing

你的選擇

你的答案: **C**

**Q22** 單選題 (2分)

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In an N-node hypercube interconnection network, the network diameter is proportional to:

- A.  $O(N)$
- B.  $O(N^2)$
- C.  $O(\sqrt{N})$
- D.  $O(\log_2(N))$

你的選擇

你的答案: **C**

**Q23 單選題** (2分)

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The concept of 'Bisection Width' in a multicomputer interconnection network evaluates:

- A. The maximum number of hops between any two nodes.
- B. The minimum number of links that must be failed to divide the network into two equal halves.
- C. The number of physical cables connected to the master node.
- D. The size of the flit buffer in a network switch.

你的選擇

你的答案: **B**

**Q24 單選題** (2分)

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A Beowulf Cluster is best described as:

- A. A tightly coupled Symmetric Multiprocessor (SMP).
- B. A chip-level multiprocessor with shared L2 cache.
- C. A distributed system utilizing the Andrew File System globally.
- D. A multicomputer built from commodity hardware, consisting of a master and slave nodes.

你的選擇

你的答案: **B**

**Q25 單選題** (2分)

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In the CBCAST (Causal Broadcast) protocol, which mechanism is fundamentally used to track and enforce the causal ordering of messages among a group of processes?

- A. A single, centralized global counter.
- B. Physical atomic hardware clocks.
- C. Vector timestamps (Vector clocks).
- D. Randomly generated sequence numbers.

你的選擇

你的答案: **D**



**Q26 單選題** (2分)

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Which of the following is the primary disadvantage of using variable-length messages instead of fixed-length messages in a message passing system?

- A. Variable-length messages cannot contain pointers.
- B. Variable-length messages consume significantly less bandwidth.
- C. They require complex dynamic memory allocation and garbage collection, causing overhead.
- D. They are fundamentally incompatible with Omega networks.

你的選擇

你的答案: C

**Q27 單選題** (2分)

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When an application continues to function normally despite the physical relocation of its required resources to a different server, the distributed system is exhibiting:

- A. Access Transparency
- B. Migration Transparency
- C. Concurrency Transparency
- D. Failure Transparency

你的選擇

你的答案: B

**Q28 單選題** (2分)

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In a multithreaded system, what is the primary advantage of a blocking receive primitive over a non-blocking receive primitive?

- A. It eliminates all network latency.
- B. It allows the thread to continuously poll the network buffer.
- C. It yields the CPU to other threads, resulting in better resource utilization.
- D. It prevents the cache coherence problem.

你的選擇

你的答案: D

**Q29** 單選題 (2分)

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Which fault tolerance technique involves executing a primary algorithm, checking the result with an Acceptance Test (AT), and rolling back to execute a secondary algorithm if the test fails?

- A. Triple Modular Redundancy (TMR)
- B. Information Redundancy
- C. Recovery Blocks (RB)
- D. Byzantine Fault Tolerance

你的選擇

你的答案: **B**

**Q30** 單選題 (2分)

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What is the defining characteristic of a 'Transient' messaging system compared to a 'Persistent' messaging system?

- A. Transient systems store messages on disk until they are explicitly deleted by the user.
- B. Transient systems discard a message if it cannot be delivered while both the sender and receiver are actively running.
- C. Transient systems guarantee delivery even if the receiver's computer is powered off.
- D. Transient systems require the use of a shared mailbox.

你的選擇

你的答案: **B**

**Q31** 單選題 (2分)

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Which type of Distributed System Middleware models the entire distributed environment as a hierarchical structure of directories and files?

- A. Document-based Middleware
- B. File System-based Middleware (e.g., Plan 9)
- C. Object-based Middleware (e.g., CORBA)
- D. Coordination-based Middleware (e.g., Linda)

你的選擇

你的答案: **D**

**Q32 單選題** (2分)

---

When using the Direct Communication Model with Asymmetric Addressing, the communication primitives are defined such that:

- A.** Both sender and receiver must name each other explicitly. 你的選擇
- B. Only the receiver must name the sender; the sender broadcasts to all.
- C. The sender names the receiver, but the receiver accepts messages from any process and returns the sender's ID.
- D. Neither party uses process IDs; they use shared mailboxes instead.

你的答案: **A**

**Q33 單選題** (2分)

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What is the key advantage of a crossbar switch?

- A. Low cost
- B. Non-blocking parallel access** 你的選擇
- C. Minimal hardware
- D. Small size

你的答案: **B**

**Q34 單選題** (2分)

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In CC-NUMA systems, cache coherence is typically maintained using:

- A. Polling
- B. Paging
- C. Directory-based protocols** 你的選擇
- D. Interrupts

你的答案: **C**

**Q35 單選題** (2分)

---

Which statement about spinlocks is MOST accurate?

- A. They are always inefficient
- B. They are best for long critical sections
- C. They waste CPU cycles but avoid context switching
- D. They work only without cache

你的選擇

你的答案: C

**Q36 單選題** (2分)

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The key difference between MP and MC is:

- A. CPU speed
- B. Memory sharing
- C. Cache size
- D. OS type

你的選擇

你的答案: D

**Q37 單選題** (2分)

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Fat trees improve performance by:

- A. Reducing nodes
- B. Increasing CPU speed
- C. Increasing bandwidth near root
- D. Removing hierarchy

你的選擇

你的答案: A

**Q38 單選題** (2分)

---

False sharing occurs when:

- A. Same variable shared
- B. Different data in same page
- C. Same CPU used
- D. Same OS

你的選擇

你的答案: B

**Q39** 單選題 (2分)

---

One-copy update semantics corresponds to:

- A. UNIX
- B. AFS
- C. HDFS
- D. NFS

你的選擇

你的答案: **D**

**Q40** 單選題 (2分)

---

Write-through caching is:

- A. Fast but unsafe
- B. Slow but reliable
- C. No disk usage
- D. Asynchronous

你的選擇

你的答案: **B**

**Q41** 單選題 (2分)

---

A replica differs from a cache because:

- A. It is temporary
- B. It is partial
- C. It is smaller
- D. It is persistent and known

你的選擇

你的答案: **A**

**Q42** 單選題 (2分)

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Quorum consensus ensures consistency when:

- A.  $r + w \leq n$
- B.  $r + w > n$  and  $2w > n$
- C.  $r = w$
- D.  $w = 1$

你的選擇

你的答案: **B**

**Q43** 單選題 (2分)

---

HDFS write policy is:

- A. Random write
- B. Write-many
- C. Read only
- D. Append only**

你的選擇

你的答案: **D**

**Q44** 單選題 (2分)

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At-least-once semantics may cause:

- A. No execution
- B. Duplicate execution**
- C. Data loss
- D. Deadlock

你的選擇

你的答案: **B**

**Q45** 單選題 (2分)

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Indirect communication uses:

- A. Registers
- B. Mailboxes
- C. CPU cache**
- D. Interrupts

你的選擇

你的答案: **C**